

GOOGLE AI IMPACT CHALLENGE 2018

Working together to apply AI for social good

[Google AI](#) conducts research that advances the state-of-the-art in the field, applying AI to products and to new domains, and developing tools to ensure that everyone can access AI.

Google.org partnered with Google AI to issue an open call to organizations around the world to submit their ideas for how they could use AI to help address societal challenges. We received applications from 119 countries, spanning 6 continents with

GRANT RECIPIENTS

Meet the Google AI Impact Challenge grantees

See how these 20 organizations are using AI to create a positive impact in their communities.



LEBANON

American University of Beirut

Millions of people in the Middle East and Africa depend on agriculture for their livelihood. Irrigation is critical to agricultural production in these arid regions, but groundwater resources are being depleted due to agricultural expansion and inefficient irrigation practices. Applying ML to weather and remotely-sensed agricultural data, this team will save water and enhance food production by providing farmers with irrigation schedules that include the precise



UNITED STATES

Crisis Text Line, Inc.

With over 100 million messages exchanged to date between people in crisis and Crisis Text Line's counselors, it can be challenging to balance spikes in volume and counselor availability. Crisis Text Line will use natural language processing and data on counselor capacity to optimize how they allocate texters to counselors, with the goal of reducing wait times while still ensuring effective communication and deescalation.

[Website](#) [Website](#) 

UGANDA

Makerere University

Air pollution is a major contributor to poor health and mortality in developing countries. Tracking spatial and temporal pollution patterns is essential to combating it, but can be difficult in low-resource environments.

Researchers from Makerere University will apply AI to data from low-cost air sensors installed on motorcycle taxis and other locations around Kampala to help improve air quality monitoring and forecasting and inform future interventions.

[Website](#) 

COLOMBIA

Colegio Mayor de Nuestra Señora del Rosario

Poor environmental practices from illegal mines pollute surrounding areas, contaminating water and fish, and endangering local workers and communities. In Colombia, researchers will use satellite imagery and machine learning to detect illegal mines throughout the country, enabling communities and the government to take actions that will protect people and natural resources.

[Website](#) 

INDONESIA

Gringgo Indonesia Foundation

Millions of tons of plastic are dumped into our oceans each year, much of it coming from communities where waste management is under-resourced or absent. Gringgo is creating an image recognition tool to help informal-sector waste collectors and independent waste management companies increase recycling rates and better integrate with city sanitation crews. This will improve and expand community trash collection and reduce ocean plastic pollution.

[Website](#) 

UNITED STATES

Quill.org

Low-income students frequently do not receive the feedback and practice needed to develop strong writing skills, and, as a result, 27 million low-income students in the United States are not proficient writers. Quill.org will use structured deep learning to provide students with immediate feedback on their writing, enabling students to revise their work and quickly improve their skills.

[Website](#) 



UNITED KINGDOM

Full Fact



UNITED STATES

Rainforest Connection

important. Together with leading fact-checking organizations AfricaCheck and Chequeado, Full Fact will use AI to provide trend monitoring and clustering tools to aid fact checkers' analysis of news and other information. This will give fact-checkers more time to focus on research, analysis, and writing articles that contextualize the news and help all of us make more informed decisions.

[Website](#) 

Connection is using commonplace mobile technology and deep learning for bioacoustic monitoring to detect immediate threats and track rainforest health. The network roots out illegal logging activity and other environmental dangers while simultaneously analyzing the health of the rainforest, making the data available for any scientist around the world.

[Website](#) 



INDIA

Wadhvani AI

More than a billion people live in smallholder farmer households worldwide, and many of these farmers struggle with avoidable pest damage that can wipe out up to 50%% of annual crop yield. In this project, as farmers and agriculture program workers take regular photos of pest traps, AI models on their phones classify and count the pests. The data will be used to provide millions of



UNITED STATES

The Pennsylvania State University

Every year, landslides cause thousands of preventable deaths around the world, and can displace entire communities and destroy habitable lands. Collaborating with the U.S. Geological Survey Landslide Hazards Program, Penn State researchers will use deep learning tools to better predict landslide location, timing, and impacted areas and create a

[Website](#) 

[Website](#) 



SWITZERLAND

HURIDOCS

Human rights lawyers are currently required to sift through vast document repositories to identify the most relevant facts for their case. HURIDOCS is using natural language processing and machine learning methods to extract, explore and connect relevant information in laws, jurisprudence, victim testimonies, and resolutions. With accessible global documents, human rights lawyers can more efficiently and effectively research and defend their cases.

[Website](#) 



BRAZIL

Hand Talk

Many deaf and hard of hearing individuals in Brazil struggle to read and write, and they rely largely on sign language to communicate. Hand Talk uses AI to translate Portuguese into Brazilian Sign Language using a digital avatar named Hugo. This tool enables digital communication for deaf and hard of hearing Brazilians, and also increases Brazilian Sign Language learning.

[Website](#) 



FRANCE

La Fondation Médecins Sans Frontières

Enabling medical staff to prescribe the right antibiotics for bacterial infections is critical for individuals and increasingly for global management of antibiotic resistance. Antibiograms help guide these decisions, but in low-resource areas, medical staff often lack the skills to interpret them. Using image recognition and a smartphone app, the Fondation Médecins Sans Frontières will help staff analyze anti-microbial images and provide appropriate patient treatment.

[Website](#)


UNITED STATES

WattTime

Billions of dollars have been spent building monitoring systems to measure the emissions of major fossil fuel power plants. These monitoring systems make critical emissions reduction initiatives possible but are not accessible in communities that cannot afford them. WattTime will use image processing algorithms and satellite networks to replace expensive, on-site power plant emissions monitors with a globally accessible, open-source monitoring platform.

[Website](#)


UNITED STATES

Nexleaf Analytics



UNITED STATES

Columbia University

...challenging in remote regions with limited infrastructure. Nexleaf tracks data on vaccine cold chain systems, providing critical information to health workers and other decision-makers on the system's functioning. They will use AI technologies to build data models that predict vaccine degradation, quantify the value of vaccines at risk, and ultimately develop an end-to-end system to ensure safe, effective vaccine delivery.

[Website](#) 

...project, researchers at Columbia University will partner with the New York City Fire Department analytics team to build models that will optimize first responder responses to the 1.7 million emergencies the department answers each year, accounting for factors such as weather, traffic, location, type of emergency, and more.

[Website](#) 



AUSTRALIA

Turning Point (Eastern Health)

Ambulances are often the first point of contact with someone who is suicidal, making ambulance clinical records a unique data source to help inform suicide prevention efforts. By using AI tools to analyze these records, Turning Point, a national center within Eastern Health, will uncover critical



UNITED STATES

The Trevor Project

In the US alone, an estimated 1.5 million LGBTQ youth seriously consider suicide or experience a significant crisis every year. The Trevor Project saves lives by supporting at-risk LGBTQ youth via phone, text, and chat. Using natural language processing and sentiment analysis, counselors will be able to determine a LGBTQ

[Website](#) 

[Website](#) 



NETHERLANDS

Skilllab BV

Helping refugees find jobs is a key priority for European cities, but translating refugees' skill sets to the European labor market is not always straightforward. Skilllab helps municipal employment services integrate refugees into local job placements. Using an app powered by AI, refugees can document their skills quickly and in their native languages, and receive recommendations of relevant career pathways to explore.

[Website](#) 



UNITED STATES

TalkingPoints

Parental involvement at home and at school is critical to students' educational outcomes, but building relationships with teachers can be challenging for families facing economic, educational, cultural or language barriers. TalkingPoints will help break down these barriers by using AI to enable two-way translated communication and personalized coaching content to guide parents' engagement with the teachers and at home with their children.

[Website](#) 

At Google, we believe that artificial intelligence can provide new ways of approaching problems and meaningfully improve people's lives. That's why we're excited to support organizations that are using the power of AI to address social and environmental challenges. Our 20 selected organizations received coaching from Google's AI experts, Google.org grant funding from a \$25 million pool, and credits and consulting from

We looked for projects across a range of social impact domains and levels of technical expertise, from organizations that are experienced in AI to those with an idea for how they could put their data to better use.

From that search, we selected 20 organizations that encompass the breadth of impact AI has to offer.

Check out the [criteria for how we selected grantees here](#), and for organizations interested in learning more about how they can get started with AI for Social Good, check out [this toolkit](#) for tips and modules.

Application criteria

Impact

How will the proposed project address a societal challenge, and to what extent? Is the application grounded in research and data about the problem and the solution? Is there a clear plan to deploy the AI model for real-world impact, and what are the expected outcomes?

Feasibility

Does the team have a well-developed, realistic plan to execute on the proposal? Does the team have a plan to access a meaningful dataset and technical expertise to apply AI to the problem? Have they identified the right partners and domain experts needed for implementation?

Use of AI

Scalability

If successful, how can this project scale beyond the initial proposal? Can it scale directly, serve as a model for other efforts, or advance the field?

Responsibility

Does the proposed use of artificial intelligence align with Google's [AI Principles](#)? See Google's [Responsible AI Practices](#) for practical guidance.

Meet the expert reviewers

Our panel of experts from around the world will help review submissions and select grant recipients.



Jeff Dean

Senior Fellow and SVP, Google AI



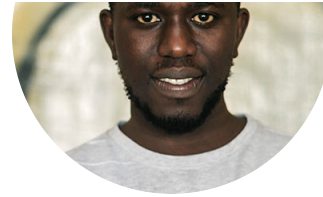
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Tim O'Reilly

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Co-founder and Head of Applied
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Fernanda Viegas

Senior Research Scientist, Google
AI

Educational resources to get started

We've developed a toolkit of educational content to help social impact organizations understand and assess how they might use AI. Whether you're a novice or interested in upleveling your ML skill set, get started by exploring our guide.

[Explore](#)

Frequently asked questions about the Challenge and how to apply.

FAQS

[Collapse all](#) 

[About](#)



What is the Google AI Impact Challenge?

The Google AI Impact Challenge is an open call to nonprofits, social enterprises, and research institutions around the world to submit their ideas to use AI to help address social and environmental challenges. Selected organizations will receive education and coaching from Google's AI experts, access to computing resources, and Google.org grant funding from a \$25M pool.

What do you mean by AI? What does it mean to apply AI for social good?

Artificial intelligence (AI) and its sub-category of machine learning (ML) are characterized by a process of training a piece of software, called a model, to generate new outputs based on a set of data inputs - for example, predicting a sports team's performance based on their past record. This model can then serve up predictions about previously unseen data. AI models can range from linear regressions run in Excel to sophisticated deep learning algorithms run on AI platforms, and we are open to applications across this spectrum, as long as they aim to help address a social or environmental problem.

Why is Google hosting the Google AI Impact Challenge?

We believe that AI can give us new ways of addressing problems and meaningfully improve people's lives. We want to support organizations on the front lines of solving today's biggest challenges with bold ideas to apply AI to the problems that they

When is the Google AI Challenge?

As of January 23, 2019, applications are now closed. We are currently reviewing proposals and will announce grant recipients in spring 2019.

What will grant recipients receive? How big will the grants be?

Grant recipients will receive the best of Google to help bring their ideas to life: coaching from Google's AI experts; Google.org grant funding from a \$25M pool; and access to resources including computing power, an ML-focused accelerator, training, and more. We expect grants may range from \$500k to \$2M USD, but will ultimately be sized based on project needs. We encourage applicants to submit budgets that accurately reflect the scope of their proposal. Likewise, education, coaching, and computing resources will also be tailored to project needs.

I applied for the program and have not heard back. How can I receive an update on the status of my organization's proposal?

Thank you for taking the time to submit a proposal to the [Google AI Impact Challenge](#). We were very impressed by the high quality of applications we received and had to make tough decisions in narrowing down the pool to the [final 20 grant recipients](#).

In the spring, we sent an application status update to the email address listed in each application. While we did email every applicant, it is possible that your organization may have had an issue receiving our email if the email address you listed is no longer valid or your organization uses filters to screen out unknown email addresses.

On behalf of the whole team at Google, we want to thank you for the time, thought and energy you put into your application, and for your continued work to make our world better.

What if I still have questions after reading the FAQs?

Entry & eligibility



Who is eligible?

The Challenge is open to any nonprofit organization, public charity, or for-profit business from around the world, provided that they apply for a project that has a charitable purpose. Unfortunately, individuals without organizational affiliation are ineligible. Public, governmental, and inter-governmental organizations are also ineligible. However, organizations that do meet our eligibility criteria and partner with governments are welcome to apply. Individuals and organizations that ordinarily reside or are located in Crimea, Cuba, Iran, North Korea, or Syria are not eligible to apply.

Can we apply if we're a for-profit business with a social impact?

For-profit businesses that are registered or formed under the laws of their country may apply if their suggested projects have an explicit charitable purpose. We're looking for innovators who understand the needs of marginalized and vulnerable populations in their own communities and around the world and are working to create new and unexpected solutions to address social or environmental inequities. Final discretion as to which organizations and which projects are funded is up to Google.

Can my organization submit a joint application with another organization?

Only one organization may be the applicant of record, but we welcome and encourage collaboration - especially between technical and social sector experts. The application allows you to specify partners who will be critical to your work. If your application is selected to receive a grant, the applicant organization will be the sole recipient of the grant, but may subgrant or subcontract with other organizations to complete the proposed project as long as all organizations comply with the terms of the grant agreement.

However, we're aiming to recruit organizations with varying levels of English ability throughout the review and selection process.

Can my organization submit more than one idea?

We encourage you to strongly consider which project best resonates with our submission criteria and your organization's strengths. However, if you feel you have two unrelated projects that would each make good candidates, please submit a separate application for each. We will review each one independently. No organization may submit more than two applications, except as designated below.

What if I work for a large organization, like a university, that has many departments - can we submit multiple applications?

Large organizations like colleges and universities are permitted to submit multiple applications. In the case of colleges and universities, we ask that each Principal Investigator submit only one proposal.

What if I am an organization that is fiscally sponsoring another?

Fiscal sponsors can submit more than one application: one for the fiscal sponsor organization itself, and one on behalf of each sponsored group.

Confidential information & IP

Will details of the project idea we submit be kept confidential?

No. Google.org will not treat your application as confidential or proprietary, and the details of your project may be shared with internal or external experts to evaluate your proposal: please do not submit any proprietary or confidential information in your application. Please keep in mind that if your organization is selected to receive a grant, your project summary will be made available to the public on the Google AI Impact Challenge website.

the general idea project supported by our funding should be able to benefit everyone. If you are selected to receive a grant, the standard grant agreement will require any intellectual property created with grant funding from Google be made available for free to the public under a permissive open source license. If your organization is selected for further consideration, and you have a compelling reason why your organization needs to own the intellectual property created with grant funding, you will have an opportunity to request an exception and provide support for your request.

Will applying to this Challenge or being selected for a grant require me to share the dataset my organization plans to use for our project with Google?

Google or its partners may ask for a small sample of your data set if you are selected for further consideration. Otherwise, Google won't need access to your organization's dataset. Please don't include your dataset or any part of it in your application.

Project info



What do you mean when you say "project"?

A project is your organization's proposed concept and implementation plan for how you will use AI to drive social good. If selected for a grant, we'll be supporting the implementation of this project - so we need to know exactly how the grant will help you realize your plan.

Do I need to have the dataset my organization plans to use for our proposed project already?

No - we will consider proposals that include a plan for how to collect or access required data, even if they have not done so yet.

Can the project be in the idea stage? Does this have to be a new idea for my organization?

need more time than we have, they may already be a year in progress. In all cases, we would like to hear exactly how a grant will change the trajectory of your progress toward implementation, scale, and impact.

Does my organization need to have experience using AI, a training dataset, or a technical expert on staff?

No - we are open to any organization with a strong idea for how to apply AI and a plan to collect or access a training dataset. We'd encourage organizations to consider partnering with technical experts in their networks for guidance and implementation support (e.g., data scientists, academic experts, or engineers).

If another organization is currently implementing a similar concept, can we still submit the idea?

Yes, but please note that projects will be judged in part on their innovative approach and potential to scale. Please tell us how and why your implementation is innovative for the community, why your organization is uniquely suited to implement the concept in a way that will be more successful, or how you plan to partner with other organizations to achieve success.

Over what time period should the grant funds be spent?

We expect the grant to be spent over the course of one to three years.

Can the grant be used to fund overhead and staffing costs?

Yes, but the large majority of the award should be devoted to the implementation of the project. For universities and other academic institutions, administrative expenses generally should be limited to 10% of the total budget or less. This maximum rate applies to the primary grantee, sub-grantees, and sub-contracts. Google.org only allows the indirect cost rates to be applied to sub-grants/contracts that are designated for research and development.

[Application process](#)



Yes, you can find a copy of the [application questions here](#).

Can we include appendices or additional information to the application?

The application includes an opportunity to provide links to additional resources. Unfortunately, we are not able to accept any attachments beyond those linked as additional materials in the application form.

Can I save a draft of my application on the site if I want to edit it later?

No. Applications can't be saved for later completion, so we recommend drafting your responses in a separate document first and only completing the application form when the entire application is ready for submission.

How do I make sure my application is successfully submitted?

Make sure all required fields are populated and within the given word limits, then click the "Submit" button. If the application has been successfully submitted, you should see a screen with a message confirming that we received your application, and you should also receive a confirmation email to the email address you provided.

I've submitted my application. What do I do now?

That's great news. No further action is required. We'll reach out if we require any additional information, and will announce the finalists per the timeline on the website.

Do I need a Google account to apply?

No, you don't need a Google account.

Will every single application be reviewed?

Yes, we will review all eligible applications received.

Share     

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		Google AI
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